# Marine non-predatory fish (sardine, anchovy, salmon, cod, herring, mackerel): HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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## Abstract

The Heavy Metal Tested & Certified standard for marine non-predatory fish, sardine, anchovy, salmon, cod, herring, mackerel, as fillet or muscle meat, fresh or frozen, stated as consumed. These lower-trophic species carry the lower government mercury ceiling; every limit is the strictest maximum level any credible government sets for the product.

## 1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default, 8:1 basis, read-across)

## 2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 200 | government maximum level |
| Cadmium (Cd) | Tier 1 | 50 | government maximum level |
| Total arsenic (tAs) | Tier 2 | 1,000 | government maximum level |
| Inorganic arsenic (iAs) | Tier 1 | 100 | government maximum level |

| | | | |
|---|---|---|---|
| Mercury, total (tHg) | Tier 1 | 300 | government maximum level |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 2,000 | human-grade read-across (European Union) |
| Tin (Sn) | Tier 2 | 200,000 | human-grade read-across (European Union) |
| Aluminium (Al) | Tier 2 | 100,000 | human-grade read-across (China) |
| Total chromium (Cr) | Tier 2 | 1,000 | government maximum level |
| Hexavalent chromium (Cr-VI) | Tier 2 | 10 | human-grade read-across (US-CA) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 200 | government maximum level |
| Cadmium (Cd) | 50 | government maximum level |
| Total arsenic (tAs) | 1,000 | government maximum level |
| Inorganic arsenic (iAs) | 100 | government maximum level |
| Mercury, total (tHg) | 300 | government maximum level |
| Methylmercury (MeHg) | ≤ tHg | bounded by parent total |
| Nickel (Ni) | 2,000 | European Union — human-food analog |
| Tin (Sn) | 200,000 | European Union — human-food analog |
| Aluminium (Al) | 100,000 | China — human-food analog |
| Total chromium (Cr) | 1,000 | government maximum level |
| Hexavalent chromium (Cr-VI) | 10 | US-CA — human-food analog |

## 4 | Cite this standard

Pendergrass, K. "Marine non-predatory fish (sardine, anchovy, salmon, cod, herring, mackerel)." In Heavy Metal Tested & Certified (HMTc) Seafood standard, version 1.0 (2026-08-07). Institute of Contaminant Standards (ICS), 2026.

# References


[1] China — human-food maximum levels
Binds Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Ni, Sn. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] US-CA — human-food maximum levels
Binds Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL



### STATUS AND LICENCE

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### COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

### DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/fish-marine--non-predatory. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

### PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/fish-marine--non-predatory.